

LEBOGANG MASENYA

Final-Year Computer Science Student — Aspiring Software Engineer and Machine Learning Engineer

Pretoria, South Africa

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SUMMARY

As a final-year computer science student with a passion for problem solving, I have always found joy in solving puzzles, whether in video games or in the world of coding. My academic journey has given me a strong foundation in statistics, full-stack web development, and machine learning, while my personal projects have allowed me to dive deeper into backend development and creating innovative systems. Beyond coding, I stay active in exercise, serve in my church, and constantly optimize my daily routines with digital planning tools.

EDUCATION

University of Pretoria

Jan 2023 – Present

Bachelor of Science in Computer Science

Pretoria, South Africa

- Key Modules: Software Engineering, Data Structures and Algorithms, Database Systems, Machine Learning, Computer Networks, Operating Systems
- Capstone: Developed a finance management application that uses AI, AR and games development

HTS Klerksdorp

Jan 2018 – Dec 2022

National Senior Certificate

Klerksdorp, South Africa

- Graduated with Distinction in Mathematics
- Modules: Computer Applications Technology, Mathematics, Physical Sciences, English Home Language

EXPERIENCE

University of Pretoria

Feb 2025 – June 2025

Student Tutor

Pretoria, South Africa

- Tutored first- and second-year Computer Science modules COS132 (Imperative Programming) and COS221 (Introduction to Databases).

VOLUNTEERING

High School Mathematics Tutor

Feb 2024 – Oct 2024

Lifting Dreams NPC

Pretoria, South Africa

- Tutored high school students in mathematics, helping them strengthen their problem-solving skills and understanding of core concepts.
- Developed strong communication skills by explaining complex mathematical ideas in a clear and accessible way.
- Provided mentorship and academic guidance to students, fostering confidence and analytical thinking.

GITHUB PROJECTS

Finance Management App

May 2025 – Present

Software Engineer — Capstone Project

Pretoria, South Africa

- Developing a full-stack application as part of my final-year Software Engineering module, collaborating with industry client Epi-Use.
- Working on AI/ML components to integrate intelligent data-driven functionalities into the system.
- Contributing as a service engineer, ensuring reliability and optimization of system performance.
- Project scope includes Augmented Reality (AR), Artificial Intelligence (AI), Machine Learning (ML), mobile development, deployment, and games development.

Shop the Look App

2024 – Present

Developer

Remote

- Developing a dynamic app that scans and isolates clothing pieces from user images.
- Integrated Google Custom Search API to retrieve online listings for fashion matches.
- Applied HuggingFace ML models for object detection and OpenCV for image pre-processing.
- Built using Flutter for mobile development, gaining full-stack experience.
- Developed skills in project planning, deployment strategies, critical thinking, and monetization fundamentals.

- Demonstrated ingenuity by leveraging limited resources to successfully achieve project goals.
- Strengthened Git workflow, debugging, and testing skills in a production environment.

Movies/Series Collection Website

Team Lead and Developer

2024
University Project

- Created a Netflix-style web application that allows users to browse, share, and save favorite content.
- Led a team of 4 developers, managing task allocation, progress tracking, and collaborative reviews.
- Focused on backend tasks: secure user authentication, session management, and database operations.
- Designed clean, responsive UI/UX to promote ease of navigation and user satisfaction.
- Learned effective team communication and conflict resolution under tight deadlines.

CERTIFICATIONS

- PHP Certificate from Jobox
- Logical Reasoning Certificate from Jobox
- Problem Solving and Analytical Thinking Certificate from Jobox
- Shopify Development from Udemy
- Pinescript financial coding from Udemy
- Univeristy of Pretoria Ready For Work Programme

SKILLS

- Programming Languages: Java, C++, Python, TypeScript
- Web and Mobile Development: Frameworks for both web and mobile apps, NodeJS, PHP
- Databases: MySQL, PostgreSQL, NoSQL, Database Development and Management
- Tools and Platforms: Git, Docker, Supabase, Microsoft Office
- Data and Analytics: Statistics, Data Analytics, Statistical Programming in R and SAS
- Machine Learning: Introductory Machine Learning Development (PyTorch), NumPY, OpenCV, Hugging Face models and Transformers
- Methodologies: Agile Development, Project Management
- Communication: Clear technical writing, professional client correspondence, and reviewer-focused documentation
- Collaboration: Cross-functional teamwork with developers, data scientists, and stakeholders
- Problem Solving: Root cause analysis, debugging complex systems, and proactive issue resolution
- Leadership: Mentoring peers, guiding architectural decisions, and presenting technical demos
- Time Management: Prioritizing tasks across development, documentation, and academic deadlines
- Critical Thinking: Evaluating trade-offs in architectural patterns and system design
- Attention to Detail: Schema integrity, trigger logic, and maintainable code practices
- Empathy: Designing systems with maintainers and end-users in mind

HONORS AND AWARDS

- **Dean's List:** Received the 2023 Dean's Merit Award in the Faculty of Engineering, Built Environment, and Information Technology.